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# Mock Dataset
import csv
import time
import random
import numpy as np

def generate_valence_mock_data(filename="valence_mock_data.csv", num_records=500):
    """
    Generates a synthetic time-series dataset for Team Valence.
    Simulates a rolling state vector paired with an observed emotional ground truth.
    """
    # Define emotional states
    states = ["Focused", "Frustrated_Masked", "Exhausted", "Collaborative_Happy"]

    # CSV Header
    header = [
        "timestamp",
        "ambient_db",
        "light_lux",
        "hours_since_break",
        "clenched_fists_count",
        "compliments_count",
        "social_withdrawal_flag",
        "facial_valence_snapshot", # The surface face label [-1.0 to 1.0]
        "true_emotional_state"     # The context-aware ground truth label
    ]

    base_time = time.time()

    with open(filename, mode='w', newline='') as file:
        writer = csv.writer(file)
        writer.writerow(header)

        for i in range(num_records):
            # 1. Simulate sequential temporal progression (sampling every 5 seconds)
            timestamp = base_time + (i * 5)

            # 2. Randomly select an underlying "room vibe" to inject realistic context correlations
            current_vibe = random.choice(states)

            # 3. Synthesize data layers based on the selected hidden room state
            if current_vibe == "Frustrated_Masked":
                ambient_db = round(random.uniform(68.0, 85.0), 1)     # Loud room
                light_lux = round(random.uniform(150.0, 300.0), 1)   # Dim room
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            hours_since_break = round(random.uniform(3.5, 6.0), 2)# Long stretch without rest
            clenched_fists_count = random.randint(1, 4)          # Somatic stress triggers
            compliments_count = 0                                # Low social reinforcement
            social_withdrawal_flag = random.choice([0, 1])
            facial_valence_snapshot = round(random.uniform(0.4, 0.8), 2) # FORCED SMILE!
(High valence)

        elif current_vibe == "Exhausted":
            ambient_db = round(random.uniform(35.0, 50.0), 1)     # Dead silent room
            light_lux = round(random.uniform(100.0, 200.0), 1)   # Very dim
            hours_since_break = round(random.uniform(5.0, 8.0), 2)# Severe exhaustion
            clenched_fists_count = 0
            compliments_count = 0
            social_withdrawal_flag = 1                           # Slumped over / isolated
            facial_valence_snapshot = round(random.uniform(-0.2, 0.1), 2) # Blank, flat face

        elif current_vibe == "Collaborative_Happy":
            ambient_db = round(random.uniform(60.0, 75.0), 1)     # Buzzing, energetic hum
            light_lux = round(random.uniform(400.0, 600.0), 1)   # Bright workspace
            hours_since_break = round(random.uniform(0.2, 1.5), 2)# Fresh off a break
            clenched_fists_count = 0
            compliments_count = random.randint(1, 3)             # Active positive reinforcement
            social_withdrawal_flag = 0
            facial_valence_snapshot = round(random.uniform(0.5, 0.9), 2) # Genuine smile

        else: # "Focused"
            ambient_db = round(random.uniform(40.0, 55.0), 1)     # Quiet focus hum
            light_lux = round(random.uniform(350.0, 500.0), 1)   # Standard light
            hours_since_break = round(random.uniform(1.0, 3.0), 2)
            clenched_fists_count = 0
            compliments_count = 0
            social_withdrawal_flag = 0
            facial_valence_snapshot = round(random.uniform(-0.4, 0.0), 2) # Intense resting focus
face (looks angry!)

        # Write row to CSV
        writer.writerow([
            timestamp, ambient_db, light_lux, hours_since_break,
            clenched_fists_count, compliments_count, social_withdrawal_flag,
            facial_valence_snapshot, current_vibe
        ])

    print(f"🎉 Successfully generated {num_records} mock records in '{filename}'!")
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if __name__ == "__main__":
    generate_valence_mock_data()
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